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Modeling and Extending RSS-based Intrusion Detection Bound via WiFi Signals

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***Abstract*—Leveraging ubiquitous WiFi infrastructures, intrusion detection methods based on Received Signal Strength (RSS) offer compelling advantages, including cost-effectiveness and privacy protection. However, existing RSS-based intrusion detection solutions fall short of accurately estimating and extending the WiFi sensing bound. In this paper, to improve intrusion detection bound, we propose a novel model of motion-disturbed RSS and design an effective R-ratio metric to extend this sensing bound. Specifically, we first establish a general model of motion-disturbed RSS and derive the blocked area and reflection area in this RSS model. Then, we define the WiFi sensing bound and propose a performance metric called R-ratio to extend the sensing bound with RSS. Furthermore, based on the statistical properties of noise, we design an efficient filter to further weaken the noise. We also provide two customized intrusion detection methods for the sensing bound. Extensive experimental results demonstrate that the proposed R-ratio-based intrusion detection method can approximately double the WiFi sensing bound compared to other methods with raw RSS data, and our developed motion-disturbed RSS model can provide valuable insights and guidance to the intrusion detection system.**

***Index Terms*—WiFi, intrusion detection, RSS, sensing bound, R-ratio metric**

I. Introduction

In recent years, with the rapid development of wireless technologies, intrusion detection gains increasing attention and has great potential in many applications, such as border protection [1], smart home [2], elderly healthcare [3]. Various techniques have been proposed for intrusion detection, such as video-based [4], infrared-based [5], sound-based [6] and sensor-based [7] approaches. By producing real-time live images, video-based approaches can create severe privacy problems for home users. Passive infrared approaches achieve better privacy protection but have limited detection angle/range. Although sound-based can support non-LoS (NLoS) scenarios, the detection accuracy is easily affected by environment noise. Sensor-based approaches attach the sensors to the body, which can be inconvenient in certain scenarios. Therefore, there is a strong desire for passive intrusion detection methods that are ubiquitous, low-cost, and prioritize privacy.

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WiFi-based intrusion detection methods [8], [9] can meet the aforementioned requirements. Up to now, a number of received WiFi signal measurements, such as Received Signal Strength (RSS) [8] and Channel State Information (CSI) [9], have been employed for passive detection of moving targets by identifying any changes. The basic principle of WiFi-based methods lies in the fact that the key features of sensitive wireless signals change with nearby intrusions. Currently, WiFi-based methods can effectively detect human motion by exploiting variations in RSS measurements [10], [11]. When connecting more smart devices, such as refrigerator, TV, and air conditioner, the RSS-based intrusion detection system can extend the monitoring area. Compared with RSS, CSI contains amplitude and phase of each subcarrier and is more sensitive to paths changes of WiFi signal, achieving a higher detection rate for human motion [12], [13].

In WiFi-based intrusion detection systems, some choose to use RSS rather than CSI for several reasons: 1) Ease of Acquisition: RSS is a relatively easy-to-obtain signal strength indicator, and almost all Wi-Fi interfaces can measure RSS. In contrast, obtaining CSI requires more complex hardware support, usually necessitating specially designed hardware, such as firmware-modified WiFi network interface cards [14]. This adds to the cost and complexity of the system. 2) Real-time Capability: Some CSI-based methods typically need to run on specific hardware platforms because they have higher performance requirements. RSS, on the other hand, is generally more real-time and can be measured on common WiFi devices, providing a quicker response for intrusion detection. 3) Computational Costs: CSI includes amplitude and phase information for each subcarrier, making it richer in information compared to RSS. However, this richness in information also increases the computational cost of processing and analyzing CSI, requiring more powerful computing resources. In some common WiFi devices with limited computing resources, using RSS is more practical. 4) Commercial Viability: The widespread use and ease of obtaining RSS make it commercially more viable. Many modern smart devices, such as smartphones and routers, support RSS measurement, while hardware support for CSI is relatively less common. Although RSS may have lower accuracy and sensitivity in certain aspects, in many application scenarios, especially where real-time requirements are high, and hardware resources are limited, choosing RSS is a more practical and feasible option.

In our daily lives, contactless sensing of wireless signals is widely acknowledged as a highly promising method. In practical applications, numerous RSS-based sensing applications necessitate prior knowledge of human presence in the

designated sensing area. However, a definitive solution to accurately determine the entry of a person into a specific area remains elusive, leading to imprecise bound sensing. This limitation constrains the practical application of RSS-based sensing in real-life scenarios. By analyzing existing RSS-based intrusion detection works, we find that existing RSS-based intrusion detection works have some limitations in calculating the sensing bound and extending the sensing range. First, in the current study, there is a lack of precise intrusion detection bound models, which hinders our accurate estimation of the sensing bound. When estimating precise sensing bounds, the following challenges are encountered: 1) Environmental complexity: In a complex and changeable environment, a wide variety of obstacles, multi-layered buildings, etc. will affect the precise sensing bounds. 2) Sensor error: Since RSS signals have problems such as multipath transmission and signal attenuation, these problems will affect the sensing bound. 3) Dynamic environment: When there are moving obstacles or people in the environment, the sensing bounds will also change accordingly. In addition, current intrusion detection methods have low accuracy, making it difficult to effectively detect human motion at long distances. When extending our sensing range, we face the following challenges: 1) Multipath interference: The increase in sensing range will lead to intensification of multipath interference, thereby increasing the complexity of intrusion detection. In addition, the extension of the sensing range will also bring noise, which will reduce the signal quality and cause false alarms, thus adversely affecting the accuracy of intrusion detection. 2) Energy efficiency: As sensor coverage increases, energy consumption also increases, especially in wireless sensor networks. Therefore, an effective energy-saving measure must be found to extend the service life of the system. 3) Data processing and analysis: Large-scale data collection and analysis also face huge challenges, which require efficient data processing algorithms and storage solutions.

To tackle the aforementioned challenges, in this paper, we explored a novel model of motion-disturbed RSS to precisely characterize the intrusion detection bound and proposed an effective R-ratio metric to extend this detection bound. The main contributions of this work are summarized as follows.

- We propose a novel model of motion-disturbed RSS for the first time to precisely characterize the intrusion detection bound. By analyzing the influence of the intruder's body size and angle on RSS and deriving the blocked area and reflection area, we successfully establish a theoretical relationship between human motion and RSS changes. Our developed motion-disturbed RSS model can provide valuable insights and guidance to the intrusion detection system.
- We propose an effective R-ratio metric to extend the detection bound. By combining RSS data at different moments, we mitigate the impact of impulse noise on amplitude. Furthermore, based on the statistical properties of noise, we design an efficient filter to further weaken the measurement noise and thereby enhance the signal. We additionally present two customized intrusion detection methods to evaluate the R-ratio performance.
- Extensive experimental results demonstrate that the proposed R-ratio-based intrusion detection method can approximately double the WiFi sensing bound compared to other methods with raw RSS data, effectively extending the sensing range from 4 to 8 meters. The experimental results further indicate that the proposed motion-disturbed RSS model can provide valuable insights and guidance to the intrusion detection system, particularly concerning the intruder's body size and angle.

The rest of this paper is organized as follow. Section II introduces the most related work. Section III presents the derivation of motion-disturbed RSS model. Analysis of broadening the sensing bound of intrusion detection is illustrated in Section IV. Then the experiment results together with the analysis is presented in Section V. Conclusions and perspectives are provided in Section VI.

II. Related Work

Previous research has employed various approaches, including label tags, wearable sensors, device-free methods [15], vision-based methods [16], or their combination [17] to detect fine-grained activities. In recent years, the utilization of WiFi signals for detecting human motion within the range of WiFi devices has gained prominence, leading to the proposal of several techniques.

A. Device-free Human Motion Detection using CSI

Qian et al. [18] extracts unsigned path length change rate information while concurrently estimating a human's moving velocity (both speed and direction) and location through an optimization method leveraging CSI amplitude signals. Hao et al. [19] successfully identified six Chinese fitness body postures in an indoor environment using CSI data. Their approach involved frequency filtering on CSI data and the application of the restricted Boltzmann machine classification algorithm, resulting in an impressive average accuracy of 85.4%. Gao et al. [20] employed CSI Ratio to effectively eliminate most CSI noise, enabling the recovery of the 2-D finger motion trajectory around devices equipped with one transmitter and two receivers. Lee et al. [21] introduced a CSI-based occupant activity monitoring system, where they successfully classified eight different activities such as bathing, cooking, eating, falling, etc., achieving an impressive average accuracy of 96%. Wang et al. [22] determined whether individuals had intruded by monitoring both the amplitude of multipath signals and changes in the angle of arrival distribution. Jia et al. [23] introduced a CSI-based fatigue detection system designed to identify drivers in potentially hazardous situations before an accident occurs. They achieved fatigue detection by analyzing a series of motions and attained an average accuracy of 89.6%. Shah et al. [24] surveilled suspicious activities in highly sensitive areas by analyzing both amplitude and phase information to ascertain the presence of intruders.

B. Device-free Human Motion Detection using RSS

Like CSI, RSS data has also been successfully deployed for human activity recognition. Based on three different change primitives in RSS caused by gesture, Abdelnasser et al. [25] constructed and recognized mutually-independent gesture families around the receiver through pattern matching. Chi et al. [26] employed RSS as the input data for their designed Human Activity Recognition (HAR) system. They achieved a notable 74% accuracy with one pair of IoT devices and an impressive 90% accuracy with three pairs in recognizing nine coarse-grained activities, including walking, sitting, running, and more. Moghaddam et al. [27] used RSS and CSI data to detect seven human-to-human interactions. They achieved 94.16% accuracy using random forest. Jin et al. [8] built RSS characteristic profiles by using non-parametric statistical method and used a joint detection scheme by using multiple data streams of WiFi node pairs to detect intrusions.

C. WiFi-based device-free sensing bound

WiFi-based device-free sensing bound detection methods have emerged these years. Zhang et al. [28] formulated a mathematical model establishing a relationship between an individual's location and the detectability of respiration within the First Fresnel Zone (FFZ). Zeng et al. [29] introduced a CSI-ratio model to tackle the issue of blind spots in respiration detection, consequently extending the sensing range. This approach required the synchronization of CSI stream clocks between two receiving antennas. Li et al. [30] derived a novel metric, DCM-CSI, from the conjugate multiplication of CSI between two antennas. Using this metric, a stable threshold line was identified to distinguish direct and indirect signals. Subsequently, the environmental walls were leveraged to establish sensing bounds.

However, existing works are all based on CSI-based sensing bound detection, and there are few RSS-based sensing bound detection methods. Therefore, our work is to precisely characterize and model the intrusion detection bound by analyzing RSS disturbed by human motion.

III. Modeling RSS-based intrusion detection bound

In this section, we first provide the definition of sensing bound. Then we build a general model of motion-disturbed RSS. To refine this RSS model and gain a more comprehensive understanding, we derive the detail expressions for both the blocked area and the reflection area within the basic model. Finally, we derive a intrusion detection bound model based on the RSS model.

A. Definition of intrusion detection bound

In WiFi-based sensing systems, the sensing range is primarily influenced by signal attenuation and propagation characteristics. The concept of the sensing bound holds significant importance as it establishes the maximum distance at which the system can reliably detect human movement. Assuming constant LoS distance for the transceiver node and an unchanged indoor environment, the farthest position where the RSS-based human intrusion detection method can detect human movement is the sensing bound.

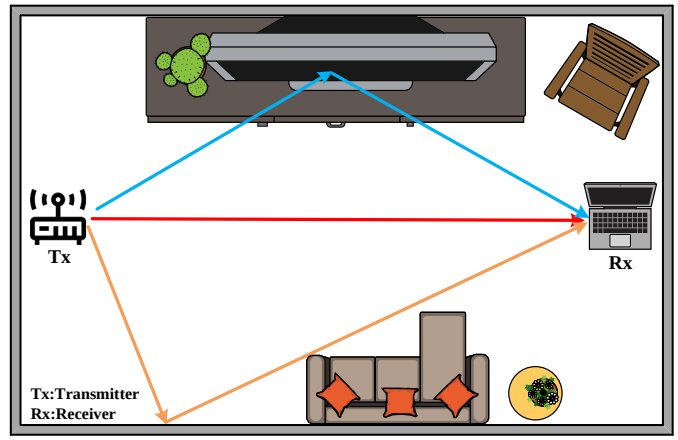


Fig. 1. Signal multipath propagation scenario.

Specifically, when the RSS features extracted by the human motion detection method are greater than the threshold, the farthest position where human motion can be detected is recorded as the intrusion detection bound. Then the sensing bound Intrusion Detection Bound (IDB) can be expressed as:

$$IDB = \max\left(d_{TP} | RSS_{ft} > RSS_{th}\right), \tag{1}$$

where d_{TP} is the distance between intruder and transceiver, RSS_{ft} is the extracted RSS feature, RSS_{th} is the threshold.

B. A general model of motion-disturbed RSS

We build a general model of motion-disturbed RSS by analyzing multipath propagation. As shown in Fig. 1, in an indoor environment, the signal arriving at the receiver is a superposition of signals coming from multiple paths. Therefore, the RSS depends on the amplitude and phase of multipath signals at the receiver and the channel frequency response (CFR) of signals at frequency f and time t can be expressed as follows:

$$H\left(f,t\right) = \sum\nolimits_{l=1}^{L} a_l\left(f,t\right) e^{\frac{-j2\pi d_l(t)}{\lambda}} + \varepsilon\left(f,t\right), \tag{2}$$

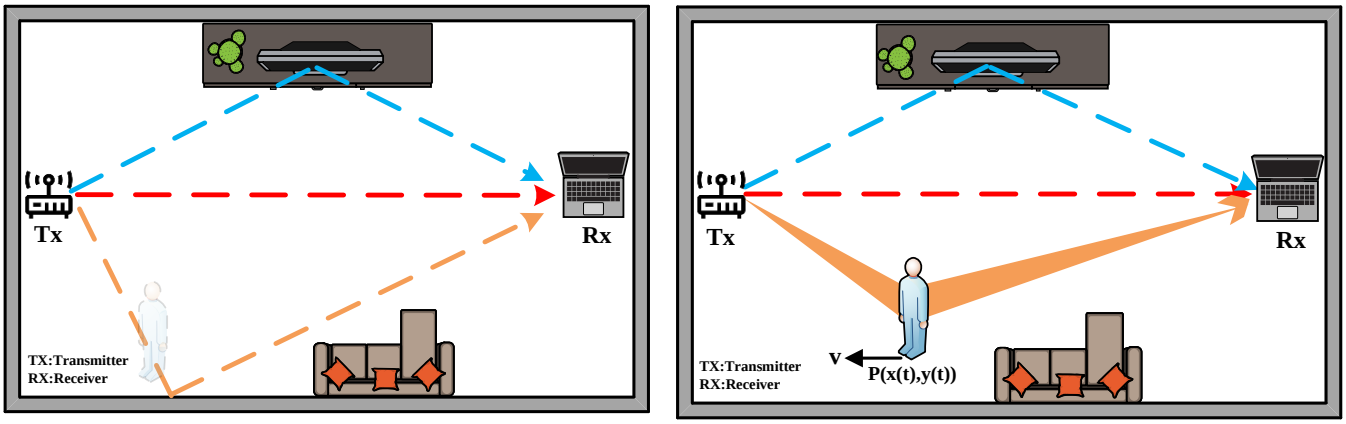
where L is the number of all paths, f is the carrier frequency, t is the time, j is imaginary number, and λ is the wavelength of the signal. $a_l\left(f,t\right)$ is the complex number representing the amplitude attenuation and initial phase of the l-th path. $d_l\left(t\right)$ is the distance of the l-th path, and $\varepsilon\left(f,t\right)$ is the measurement noise. The transmitted power is denoted by P_t, then the RSS can be expressed as follows:

$$RSS\left(f,t\right) = \left|H\left(f,t\right)\right|^2 \times P_t. \tag{3}$$

Combining equations (2) and (3), we have

$$RSS\left(f,t\right) = \left|\sum\nolimits_{l=1}^{L} a_l\left(f,t\right) e^{\frac{-j2\pi d_l(t)}{\lambda}} + \varepsilon\left(f,t\right)\right|^2 \times P_t, \tag{4}$$

where $L = 1$ represents the shortest path. If the line of sight (LoS) path exists, the shortest path is the LoS path. Otherwise, the shortest path is the shortest non-line of sight (NLoS) path.



(a) No-intruder scenario. (b) Intruder scenario.

Fig. 2. Signal propagation.

The complex coefficient $a_l(f,t)$ is related to the propagation distance $d_l(t)$ and can be expressed as:

$$a_l(f,t) = \frac{\sqrt{G_l}\lambda}{4\pi}\left[\frac{1}{d_l(t)}\right]^{\frac{n}{2}} \cdot e^{j\Phi_l(f,t)}, \tag{5}$$

where G_l is the product of the antenna gain at the transmitter and the receiver, n is the path loss factor, $\Phi_l(f,t)$ is the initial phase of the l-th path.

We assume that there is only one intruder in an indoor environment. When the intruder appears around transceiver devices, $H(f,t)$ is typically the superposition of static component and dynamic component, and $H(f,t)$ is affected by noise. Therefore, $H(f,t)$ can be written as:

$$H(f,t) = A_{noise}e^{-j\theta_{offset}(f,t)}\left(H_s(f) + H_d(f,t) + \varepsilon(f,t)\right), \tag{6}$$

where A_{noise} is the impulse noise in amplitude and $e^{-j\theta_{offset}(f,t)}$ is the random phase offset. $H_s(f)$ is static component and remains constant within a short period of time, and $H_d(f,t)$ is dynamic component. $H_s(f)$ corresponds to static signals, including the LoS path signal (if LoS path exists) and the signals reflected by static objects such as walls and floors. $H_d(f,t)$ corresponds to dynamic signals which are reflected by moving objects such as an intruder and change with human movements.

As illustrated in Fig. 2(a), when there is no person in an indoor environment, the static signal arriving at the receiver is the superposition of signals coming from all static paths. In the multipath-rich environment, the shortest path is directly from the transmitter to the receiver if the LoS propagation is present. Other paths comprise reflective paths on which the transmitted signal is reflected once or more times at various kinds of reflectors such as the walls and floors. Each reflection generates corresponding reflection loss, and the value of reflection loss is related to the material of the reflection surface. Therefore, $H_s(f)$ can be expressed as follows:

$$H_s(f) = \sum\nolimits_{k\in P_s}\prod\nolimits_{n=1}^{N_k} S_k^{(n)}\Gamma_k^{(n)}a_k(f)\,e^{\frac{-j2\pi d_k}{\lambda}} + \varepsilon_s(f), \tag{7}$$

where P_s is the set of all static paths, $a_k(f)$ is the complex number representing the amplitude attenuation and initial phase of the k-th path, and d_k is the distance of the k-th path. N_k is the number of reflections of the k-th path, and $S_k^{(n)}$ is the reflection area of n-th reflection of the k-th path. $\Gamma_k^{(n)}$ is the reflection loss coefficient per unit reflection area of n-th reflection of the k-th path, and $\varepsilon_s(f)$ is the measurement noise in a static environment. According to equation (7), multiple reflections dramatically weaken the signal arriving at the receiver. In this paper, we only consider once-reflection path, so equation (7) can be abbreviated as:

$$H_s(f) = \sum\nolimits_{k\in P_s} S_k\Gamma_k a_k(f)\,e^{\frac{-j2\pi d_k}{\lambda}} + \varepsilon_s(f), \tag{8}$$

where S_k is the reflection area of once reflection of the k-th path, and Γ_k is the reflection loss coefficient per unit reflection area of once reflection of the k-th path.

As shown in Fig. 2(b), when someone appears in an indoor environment, the RSS is disturbed. Because some static paths are blocked by the human body and these paths change their reflectors to the human body. $H_d(f,t)$ corresponds to signals which are reflected and blocked by the human body and can be written as:

$$\begin{aligned} H_d(f,t) = {} & S_{mo}(t)\,\Gamma_{mo}a_{mo}(f,t)\,e^{\frac{-j2\pi d_{mo}(t)}{\lambda}} - \\ & S_{bl}(t)\,\Gamma_{bl}a_{bl}(f,t)\,e^{\frac{-j2\pi d_{bl}(t)}{\lambda}} + \varepsilon_d(f,t), \end{aligned} \tag{9}$$

where $S_{mo}(t)$ is the area reflected by the human body, $S_{bl}(t)$ is the area blocked by the human body, Γ_{mo} is the reflection coefficient per unit area reflected by the human body, and Γ_{bl} is the reflection coefficient per unit area blocked by the human body. $a_{mo}(f,t)$ and $a_{bl}(f,t)$ are complex numbers representing the amplitude attenuation and initial phase of the path reflected by the human body and the path blocked by the human body, respectively. $d_{mo}(t)$ and $d_{bl}(t)$ are the distance of the path reflected by the human body and the distance of the path reflected by the human body, respectively. And $\varepsilon_d(f,t)$ is the measurement noise in a human environment.

Since static path signals in the indoor environment remain constant over a short period of time. Combining equations (3),

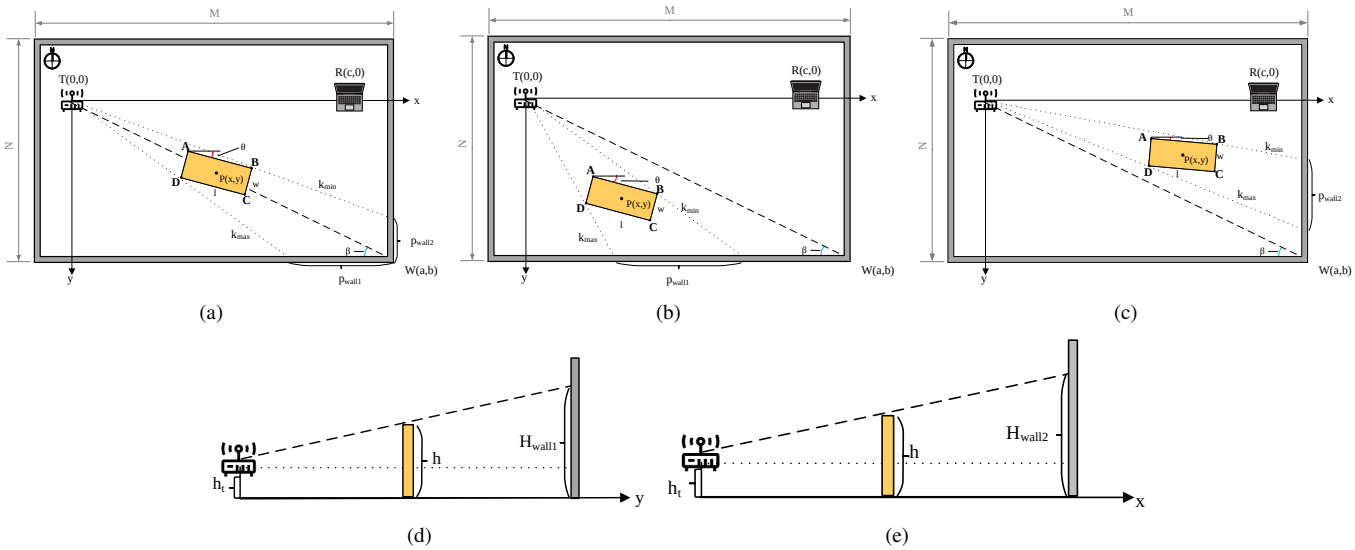


Fig. 3. The projection of the transmitted signal blocked by the human body on the wall:(a) Top view projected on both walls, (b) Top view projected only on south wall, (c) Top view projected only on east wall, (d) Side view projected on the south wall, (e) Side view projected on the east wall.

(6), and (9), RSS can be written as:

$$RSS\,(f,t) = |A_{noise}e^{-j\theta_{offset}(f,t)}(H_s(f)+ S_{mo}(t)\Gamma_{mo}a_{mo}(f,t)e^{\frac{-j2\pi d_{mo}(t)}{\lambda}}- S_{bl}(t)\Gamma_{bl}a_{bl}(f,t)e^{\frac{-j2\pi d_{bl}(t)}{\lambda}}+\varepsilon_d(f,t))|^2\times P_t. \tag{10}$$

Observing from the equation (10), RSS increases when the area blocked by the human body decreases, conversely, the RSS decreases when the area blocked by the human body increases. Similarly, when the area reflected by the human body decreases, the RSS decreases, and when the area reflected by the human body increases, the RSS increases. We can observe that both the blocked area and reflection area have a significant on RSS. In the next subsection, we will derive the blocked area and reflection area in detail.

C. Derivation of the blocked area in RSS model

Based on the equation (10), S_{bl} represents the area blocked by the human body, which changes with the orientation and position of the human body. We approximate the area blocked by the human body by calculating the projected area on the wall. We can know that the length of the room is M and the width is N. The transmitter is located at distances a and b from the east and south walls, respectively. The distance between the transmitter and the receiver is c, and the connecting line between the transmitter and the receiver is parallel to the south wall. To establish a two-dimensional coordinate system, we consider the transmitter as the origin. We designate the east direction as the positive x-axis direction and the south direction as the positive y-axis direction. The x-axis is aligned parallel to the south wall. We model the human body as a cube with dimensions of length, width, and height, denoted as l, w, and h, respectively. As shown in Fig. 3, the angle between the line segment TW and the south wall is β, the angle between the front of the human body and the south wall is θ, and the clockwise direction of the angle is defined as the positive direction. The person is located at point $P(x,y)$, and the shape of the person when viewed from above is a rectangle of length l and width w, with the four vertices denoted as A, B, C, and D, respectively.

We can easily calculate the coordinates of the four vertices. Subsequently, we calculate the slopes between the transmitter and these vertices to determine the boundaries of the blocked range. The range blocked by the human body can be determined by considering the maximum slope denoted as $k_{\max}$ and the minimum slope denoted as $k_{\min}$. These two slopes correspond to the left and right boundaries of the blocked range, respectively. As illustrated in Fig. 3(a), Fig. 3(b) and Fig. 3(c), when a person is in motion, the transmitted signals blocked by the human body are projected only on the south wall or only on the east wall or on both walls. The projected height on the south wall and the east wall are illustrated in Fig. 3(d) and Fig. 3(e). According to the triangle proportionality property, the blocked area projected on the wall can be expressed as follows:

$$S_{bl} = S_{bl1} + S_{bl2}, \tag{11}$$

where

$$\begin{aligned} S_{bl1} &= P_{wall1}\times H_{wall1} \\ &= bmax\left(\frac{1}{max(k_{\min},\tan\beta)}-\frac{1}{k_{\max}},0\right)\left(\frac{b(h-h_t)}{y}+h_t\right), \end{aligned} \tag{12}$$

$$\begin{aligned} S_{bl2} &= P_{wall2}\times H_{wall2} \\ &= amax\left(\min\left(k_{\max},\tan\beta\right)-k_{\min},0\right)\left(\frac{a(h-h_t)}{x}+h_t\right), \end{aligned} \tag{13}$$

with $k_{\max} = \max\left(k_1,k_2,k_3,k_4\right)$, $k_{\min} = \min\left(k_1,k_2,k_3,k_4\right)$ and $\tan\beta = \frac{b}{a}$. Then, we have $k_1 = \frac{y-l/2\sin\theta-w/2\cos\theta}{x+w/2\sin\theta-l/2\cos\theta}$, $k_2 = \frac{y+l/2\sin\theta-w/2\cos\theta}{x+w/2\sin\theta+l/2\cos\theta}$, $k_3 = \frac{y+l/2\sin\theta+w/2\cos\theta}{x-w/2\sin\theta+l/2\cos\theta}$, and $k_4 = \frac{y-l/2\sin\theta+w/2\cos\theta}{x-w/2\sin\theta-l/2\cos\theta}$. When $\frac{1}{max(k_{\min},\tan\beta)}-\frac{1}{k_{\max}}\leq 0$, then

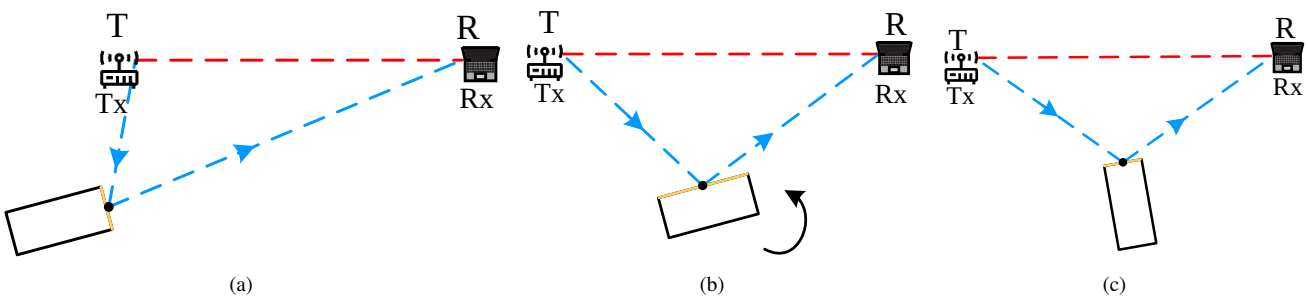


Fig. 4. Reflection area of the human body at different positions and orientations.

$S_{bl} = S_{bl2}$. At this time, the blocked area is only projected on the east wall, as shown in Fig. 3(e). When $\min(k_{\max}, \tan\beta) - k_{\min} \leq 0$, $S_{bl2} = 0$, then $S_{bl} = S_{bl1}$. At this time, the blocked area is only projected on the south wall, as shown in Fig. 3(d).

D. Derivation of the reflection area in RSS model

As observed in Fig. 4(a) and Fig. 4(b) that the reflection area varies when a person is in the same orientation but at different locations. Similarly, when a person is at the same location but in different orientations, the reflection area is also different, as shown in Fig. 4(b) and Fig. 4(c). The orientations and locations of the human body constantly change during movement. Next, we introduce how to calculate the effective reflection area of the human body.

Fig. 5 displays a top view of the connection between the central points of the four surfaces of the human body and the transmitter and receiver. The human body is regarded as a rectangle, with four vertices marked as E, F, G, and H, respectively. In addition, S_f, S_b, S_l, and S_r are the central points of the front, back, left and right surfaces of the human body, respectively.

Based on the laws of reflection and refraction of electromagnetic waves, when electromagnetic waves reach the human body surface, electromagnetic waves are reflected. For the paths TS_lR, TS_rR, and TS_bR, electromagnetic waves need to penetrate the human body to reach the receiver. However, penetrating the human body is very difficult due to the fact that approximately 70% of the human body is composed of water, and electromagnetic waves experience significant attenuation in water. Therefore, the surfaces of the human body where the points S_b, S_l, and S_r are located cannot be regarded as reflection surfaces. For the path TS_fR, the signal can reach the receiver without penetrating the human body only through one reflection. Thus, the surface of the human body where the point TS_fR is located can be regarded as the effective reflection surface and its area is the effective reflection area.

Based on the above analysis, in this paper, the determination of the effective reflection surface depends on whether the line between the central point of a particular surface of the human body and the transceiver passes through the human body. Take the right surface of the human body (i.e.FG surface) for example, if line segments TS_r or RS_r intersect with line segments EF, EH, or GH, the FG surface cannot be regarded as the effective reflection surface; if not, the FG

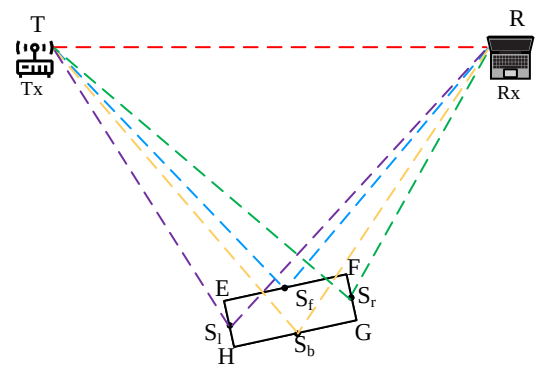


Fig. 5. Top view of the connection between the central points of the four surfaces of the human body and the transmitter and receiver.

surface is regarded as the effective reflection surface. The intersection of two line segments is determined by computing a vector cross-product. Take the line segment TS_r and line segment EF for example,as illustrated in Fig. 6. Points E and F are on opposite sides of line segment TS_r, so vectors $\overrightarrow{TE}$ and $\overrightarrow{TF}$ are on opposite sides of vector TS_r resulting in cross products with different signs. Similarly, points T andS_r are on opposite sides of line segment EF, so vectors $\overrightarrow{ET}$ and $\overrightarrow{ES_r}$ are on opposite sides of vector $\overrightarrow{EF}$, resulting in cross products with different signs. When both conditions are simultaneously met, the two line segments intersect. We define $I(Line1, Line2)$ to represent the result of the intersection of two line segments, we have

$$I(TS_b, EF) = \begin{cases} 1, if \begin{cases} \left(\overrightarrow{ET} \times \overrightarrow{EF}\right) \cdot \left(\overrightarrow{ES_b} \times \overrightarrow{EF}\right) \leq 0 \\ \left(\overrightarrow{TE} \times \overrightarrow{TS_b}\right) \cdot \left(\overrightarrow{TF} \times \overrightarrow{TS_b}\right) \leq 0 \end{cases} \\ 0, otherwise \end{cases} \tag{14}$$

Furthermore, we use $J(Surface)$ to represent whether a particular surface of the human body can be regarded as the reflection surface, for the back surface of the human body (i.e.FG surface), we have

$$J(FG) = \sum_{Center \in \{TS_r, RS_r\}} \sum_{Surface \in \{EF, EH, GH\}} I(Center, Surface), \tag{15}$$

where $Center$ is the set of the lines between the transceiver and the central point of the FG surface, and $Surface$ is the set

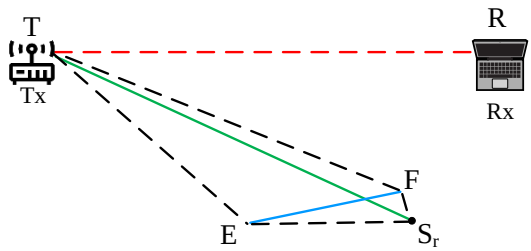


Fig. 6. Top view of the connection between the central point of the right surface of the human body and the transmitter.

of the remaining three surfaces. If $J(GH) = 0$, representing there is no intersection between line segments TS_b and RS_b and line segments EF, EH, and FG, the GH surface is regarded as the reflection surface. If $J(GH) \neq 0$, the GH surface is not the reflection surface. The same is true for other surfaces. In summary, the effective reflection area of the human body can be expressed as:

$$S_{mo} = \begin{cases} l \times h, if J(EF) = 0 \parallel J(GH) = 0 \\ w \times h, if J(EH) = 0 \parallel J(FG) = 0 \\ 0, otherwise \end{cases} . \quad (16)$$

At this point, we have successfully derived the blocked area S_{bl} and the reflection area S_{mo}. Substituting them into equation (10), we can get the complete model. Our analysis shows that S_{bl} varies with position and orientation of human during human movement. Similarly, due to different positions and orientations, the reflective surface on the human body is different, resulting in different reflection area S_{mo}.

E. Derivation of RSS-based intrusion detection bound model

According to the definition of intrusion detection bound in equation(1), when the RSS features extracted by human motion detection method are greater than the threshold, the farthest position where human motion can be detected is recorded as the intrusion detection bound. To obtain the IDE, we use two different intrusion detection methods: improved FFT method based on maximum amplitude and improved wavelet transform method based on scale projection.

When there is no intruder, the measured RSS value is recorded as $rss_0(t)$; when an intruder appears, the measured RSS value is recorded as $rss_d(t)$. $rss_0(t)$ and $rss_d(t)$ can be obtained according to equation(10). For the improved FFT method based on maximum amplitude, we first perform FFT on RSS data measured in static environment and intrusion environment respectively:

$$\mathbf{FFT_s}(m) = \sum_{t=0}^{N-1} rss_0(t)\, e^{\frac{-j2\pi mt}{N}}, m \in [0, N-1], \quad (17)$$

$$\mathbf{FFT_d}(n) = \sum_{t=0}^{N-1} rss_d(t)\, e^{\frac{-j2\pi nt}{N}}, n \in [0, N-1], \quad (18)$$

where $rss_0(t) = \left|A_{noise} e^{-j\theta_{offset}(f,t)} (H_s(f) + \varepsilon_d(f,t))\right|^2 \times P_t$, and $rss_d(t) = |A_{noise} e^{-j\theta_{offset}(f,t)} (H_s(f) + S_{mo}(t)\Gamma_{mo} a_{mo}(f,t)\, e^{\frac{-j2\pi d_{mo}(t)}{\lambda}} - S_{bl}(t)\Gamma_{bl} a_{bl}(f,t)\, e^{\frac{-j2\pi d_{bl}(t)}{\lambda}} + \varepsilon_d(f,t))|^2 \times P_t$. $\mathbf{FFT_s}(m)$ represents the m-th output result after fast Fourier transform, $\mathbf{FFT_d}(n)$ represents the n-th output result after fast Fourier transform, and N is the number of signal samples. Then we extract the maximum amplitude except the 0-frequency component: $M_s = \max(|\mathbf{FFT_s}[1:N-1]|)$, $M_d = \max(|\mathbf{FFT_d}[1:N-1]|)$. As the intruder moves further away from WiFi devices, their impact on wireless channels decreases and M_d gradually decreases. When M_d becomes small enough, it is not possible to perceive the intrusion of the intruder. At this point, the threshold RSS_{th} of (1) is set to M_s.

For the improved wavelet transform method based on scale projection, we first perform DWT on RSS data measured in static environment and intrusion environment respectively and and extract detail coefficients:

$$\mathbf{DWT_s}_i^{(l)} = \sum_{t \in Z} rss_0(t)\, h_{t-2^l i}^{(l)}, \quad (19)$$

$$\mathbf{DWT_d}_k^{(j)} = \sum_{t \in Z} rss_d(t)\, h_{t-2^j k}^{(j)}, \quad (20)$$

where l and j represent the decomposition level of DWT respectively, $h^{'}$s represent two sets of discrete orthogonal functions. After obtaining the detail coefficients, we calculate the corresponding variance: $V_s = \mathrm{var}(\mathbf{DWT_s})$, $V_d = \mathrm{var}(\mathbf{DWT_d})$. Similarly, as the intruder moves further away from WiFi devices, their impact on wireless channels decreases and V_d gradually decreases. When V_d becomes small enough, it is not possible to perceive the intrusion of the intruder. At this point, the threshold RSS_{th} of (1) is set to V_s .

We fix the receiver and transmitter, and the intruder gradually moves away from the WiFi link. The intruder moves at different distances from the WiFi link, during which the receiver collects RSS data. Obtain RSS_{ft} through the above method, continuously check if the constraint conditions are met. If they are met, the intruder will continue to stay away from the WiFi link and repeat the above steps. When the detected result does not meet the constraint conditions, the corresponding d_{TP} is the IDE. According to equation(10), the intrusion detection bound model is closely related to the blocked area and reflection area. During the intrusion process, according to equation(11), when calculating the blocked area, the projected area may be on both the east and south walls, or in specific situations, only on the east or south wall. Consequently, the corresponding equation for calculating the blocked area varies, thereby affecting the establishment of the intrusion detection bound model. Similarly, in accordance with equation(16), the reflection surface of a person may be on the front, left side, right side, or lateral side, leading to diverse equations for calculating the reflection area and, in turn, resulting in different intrusion detection bound models.

During this process, due to factors such as multipath and noise affecting RSS data, the intrusion detection bound is small. Therefore, we have designed an effective method to expand the intrusion detection bound. The specific process will be discussed in Section IV.

IV. Broadening the sensing bound of intrusion detection

In this section, we first propose a performance metric called R-ratio to extend the sensing bound with RSS, and then give two detection methods for the sensing bound.

A. Extending the sensing bound

Expanding $|H(f,t)|^2$ in equation (10), we can get :

$$\begin{aligned}|H(f,t)|^2 = {A_{noise}}^2(|a_s(f)|^2 + |S_mo(f,t)|^2 \\ +2\,|a_s(f)S_mo(f,t)|\cos(\phi_{bls}(t)) \\ -2\,|a_s(f)S_bl(f,t)|\cos(\phi_{bls}(t)) \\ +2\,|a_s(f)a_\varepsilon(f,t)|\cos(\Delta\phi_{\varepsilon s}) \\ -2\,|S_bl(f,t)S_mo(f,t)|\cos(\phi_{blmo}(t)) \\ +2\,|a_\varepsilon(f,t)S_mo(f,t)|\cos(\phi_{mo\varepsilon}(t)) \\ -2\,|a_\varepsilon(f,t)S_bl(f,t)|\cos(\phi_{bl\varepsilon}(t))),\end{aligned} \tag{21}$$

where $S_bl(f,t) = S_{bl}(t)\,\Gamma_{bl}a_{bl}(f,t)$ corresponds to the dynamic component which blocked by human body. $S_mo(f,t) = S_{mo}(t)\,\Gamma_{mo}a_{mo}(f,t)$ corresponds to the dynamic component which reflected by human body. We have also $\cos(\phi_{bls}(t)) = \cos\left(\frac{2\pi d_{bl}(t)}{\lambda} - \phi_s\right)$, $\cos(\phi_{mos}(t)) = \cos\left(\frac{2\pi d_{mo}(t)}{\lambda} - \phi_s\right)$, $\cos(\Delta\phi_{\varepsilon s}) = \cos(\phi_\varepsilon - \phi_s)$, $\cos(\phi_{blmo}(t)) = \cos\left(\frac{2\pi d_{mo}(t) - 2\pi d_{bl}(t)}{\lambda}\right)$, $\cos(\phi_{bl\varepsilon}(t)) = \cos\left(\frac{2\pi d_{bl}(t)}{\lambda} - \phi_\varepsilon\right)$, and $\cos(\phi_{mo\varepsilon}(t)) = \cos\left(\frac{2\pi d_{mo}(t)}{\lambda} - \phi_\varepsilon\right)$.

In reality, apart from measurement noise, RSS readings in commodity WiFi devices contain severe noise comprising of the impulse noise in amplitude, and random offset in phase. All of these factors contribute to the RSS variation caused by motion. We assume that impulse noise remains almost constant in amplitude within a short period of time.

To alleviate the impact of pulse noise on the amplitude, we subtract the RSS value at time t from the RSS value at time t_0 and divide the result by the value at time t_0. Then, the specific mathematical expression is as follows:

$$R = \frac{|H(f,t)|^2 \times P_t - |H(f,t_0)|^2 \times P_t}{|H(f,t_0)|^2 \times P_t} = \frac{|H(f,t)|^2}{|H(f,t_0)|^2} - 1. \tag{22}$$

The R of equation (22), represents a normalized measure of the change in RSS over time. Substituting equation (21) into equation (22), we can get equation (23).

In our experiments, we consider a relatively stable indoor environment in which the building structure and object layout are relatively unchanged. This scenario is usually common in real life, especially in a home environment where owners are typically absent during working hours. In this scenario, it can be considered that the relatively static building structure and object layout will dominate the static components in the intrusion detection system, while the relatively small dynamic components and noise components may be more in line with the actual situation. Therefore, the noise component and dynamic component are much smaller than the static component. We can reasonably ignore the ②, ③, ⑤, ⑦, ⑧, ⑩ components in the numerator and denominator. Then, equation (23) can be rewritten as:

$$\begin{aligned}R \approx \Big(|a_s(f)|^2 + 2\,|a_s(f)|\,\Big(|S_mo(f,t)|\cos(\phi_{mos}(t)) - \\ |S_bl(f,t)|\cos(\phi_{bls}(t)) + |a_\varepsilon(f,t)|\cos(\Delta\phi_{\varepsilon s})\Big)\Big) \Big/ \\ \Big(|a_s(f)|^2 + 2\,|a_s(f)|\,\Big(|S_mo(f,t_0)|\cos(\phi_{mos}(t_0)) - \\ |S_bl(f,t_0)|\cos(\phi_{bls}(t_0)) + |a_\varepsilon(f,t_0)|\cos(\Delta\phi_{\varepsilon s})\Big)\Big) - 1.\end{aligned} \tag{24}$$

Then, we $|a_s(f)|^2$ extract in the denominator of equation (24) to prepare for its subsequent transformation into the form of Taylor series expansion, equation (24) can be rewritten as:

$$\begin{aligned}R \approx \Big(|a_s(f)|^2 + 2\,|a_s(f)|\,\Big(|S_mo(f,t)|\cos(\phi_{mos}(t)) - \\ |S_bl(f,t)|\cos(\phi_{bls}(t)) + |a_\varepsilon(f,t)|\cos(\Delta\phi_{\varepsilon s})\Big)\Big) \Big/ \\ \Big(|a_s(f)|^2\Big(1 + 2\,|a_s(f)|\,\Big(|S_mo(f,t_0)|\cos(\phi_{mos}(t_0)) - \\ |S_bl(f,t_0)|\cos(\phi_{bls}(t_0)) + |a_\varepsilon(f,t_0)|\cos(\Delta\phi_{\varepsilon s})\Big) \Big/ \\ |a_s(f)|^2\Big)\Big) - 1.\end{aligned} \tag{25}$$

In our experiment, we assume that there is no intrusion at time t_0 , so there is no dynamic component at time t_0, then equation (25) can be rewritten as:

$$\begin{aligned}R \approx \Big(|a_s(f)|^2 + 2\,|a_s(f)|\,\Big(|S_mo(f,t)|\cos(\phi_{mos}(t)) - \\ |S_bl(f,t)|\cos(\phi_{bls}(t)) + |a_\varepsilon(f,t)|\cos(\Delta\phi_{\varepsilon s})\Big)\Big) \Big/ \\ |a_s(f)|^2\Big(1 + 2\,|a_s(f)a_\varepsilon(f,t_0)|\cos(\Delta\phi_{\varepsilon s}) \Big/ |a_s(f)|^2\Big) - 1.\end{aligned} \tag{26}$$

Since the noise component is smaller than the static component, we can reasonably assume that $2\,|a_s(f)\,a_\varepsilon(f,t_0)|\cos(\Delta\phi_{\varepsilon s})$ is much smaller than $|a_s(f)|^2$. Therefore, we can appropriately truncate the Taylor series expansion :

$$\begin{aligned}&\frac{1}{1 + \frac{2|a_s(f)a_s(f,t_0)|\cos(\Delta\phi_{\varepsilon s})}{|a_s(f)|^2}} \\ &\approx \sum_{k=0}^{\infty}(-1)^k\left(\frac{2\,|a_s(f)\,a_s(f,t_0)|\cos(\Delta\phi_{\varepsilon s})}{|a_s(f)|^2}\right)^k,\end{aligned} \tag{27}$$

and we keep only the first two terms, equation (27) can be rewritten as:

$$\begin{aligned}&\frac{1}{1 + \frac{2|a_s(f)a_\varepsilon(f,t_0)|\cos(\Delta\phi_{\varepsilon s})}{|a_s(f)|^2}} \\ &\approx 1 - \frac{2\,|a_s(f)\,a_\varepsilon(f,t_0)|\cos(\Delta\phi_{\varepsilon s})}{|a_s(f)|^2}.\end{aligned} \tag{28}$$

The dynamic component and noise component are smaller than the static component, so $2\,|a_s(f)\,S_mo(f,t)|\cos(\phi_{mos}(t))$ and $2\,|a_s(f)\,a_\varepsilon(f,t)|\cos(\Delta\phi_{\varepsilon s})$ are smaller than $|a_s(f)|^2$. Put equation (28) into equation (26), we can obtain:

$$
\begin{aligned}
R = \Bigg(& \underbrace{|a_s(f)|^2}_{①} + \underbrace{|S_mo(f,t)|^2 + |S_bl(f,t)|^2 + |a_\varepsilon(f,t)|^2}_{②} - \underbrace{2|S_bl(f,t)\,S_mo(f,t)|\cos(\phi_{blmo}(t))}_{③} + \\
& \underbrace{2|a_s(f)|\left(|S_mo(f,t)|\cos(\phi_{mos}(t)) - |S_bl(f,t)|\cos(\phi_{bls}(t)) + |a_\varepsilon(f,t)|\cos(\Delta\phi_{\varepsilon s})\right)}_{④} + \\
& \underbrace{2|a_\varepsilon(f,t)|\left(|S_mo(f,t)|\cos(\phi_{mo\varepsilon}(t)) - |S_bl(f,t)|\cos(\phi_{bl\varepsilon}(t))\right)}_{⑤}\Bigg) \Bigg/ \\
\Bigg(& \underbrace{|a_s(f)|^2}_{⑥} + \underbrace{|S_mo(f,t_0)|^2 + |S_bl(f,t_0)|^2 + |a_\varepsilon(f,t_0)|^2}_{⑦} - \underbrace{2|S_bl(f,t_0)\,S_mo(f,t_0)|\cos(\phi_{blmo}(t_0))}_{⑧} + \\
& \underbrace{2|a_s(f)|\left(|S_mo(f,t_0)|\cos(\phi_{mos}(t_0)) - |S_bl(f,t_0)|\cos(\phi_{bls}(t_0)) + |a_\varepsilon(f,t_0)|\cos(\Delta\phi_{\varepsilon s})\right)}_{⑨} + \\
& \underbrace{2|a_\varepsilon(f,t_0)|\left(|S_mo(f,t_0)|\cos(\phi_{mo\varepsilon}(t_0)) - |S_bl(f,t_0)|\cos(\phi_{bl\varepsilon}(t_0))\right)}_{⑩}\Bigg) - 1.
\end{aligned}
\tag{23}
$$

$$
\begin{aligned}
R \approx & \frac{2|S_mo(f,t)|\cos(\phi_{mos}(t))}{|a_s(f)|} - \frac{2|S_bl(f,t)|\cos(\phi_{bls}(t))}{|a_s(f)|} \\
& + \frac{2|a_\varepsilon(f,t)|\cos(\Delta\phi_{\varepsilon s})}{|a_s(f)|} - \frac{|a_s(f)|}{|a_s(f)|}\frac{2|a_\varepsilon(f,t_0)|\cos(\Delta\phi_{\varepsilon s})}{|a_s(f)|}.
\end{aligned}
\tag{29}
$$

We mark $\frac{2|a_\varepsilon(f,t)|\cos(\Delta\phi_{\varepsilon s})}{|a_s(f)|} - \frac{2|a_\varepsilon(f,t_0)|\cos(\Delta\phi_{\varepsilon s})}{|a_s(f)|}$ as $\frac{|a_\varepsilon(f,t)'|}{|a_s(f)|}$, so equation (29) can be rewritten as:

$$
\begin{aligned}
R \approx & \underbrace{\frac{2|S_mo(f,t)|\cos(\phi_{mos}(t))}{|a_s(f)|}}_{\mathrm{I}} - \\
& \underbrace{\frac{2|S_bl(f,t)|\cos(\phi_{bls}(t))}{|a_s(f)|}}_{\mathrm{II}} + \underbrace{\frac{|a_\varepsilon(f,t)'|}{|a_s(f)|}}_{\mathrm{III}}.
\end{aligned}
\tag{30}
$$

The newly constructed RSS ratio signal has three parts: I corresponds to the dynamic part of human body reflection; II corresponds to the dynamic part of human body blocked; III corresponds to noise.

We define R-ratio to represent the logarithmic transformation of R (in dBm). Using dBm as the unit of measurement is more helpful for visually comparing changes in RSS. The defined R-ratio is expressed as follows:

$$
\mathrm{R-ratio\,(dBm)} = 10\log\left(\frac{R}{1\mathrm{mW}}\right). \tag{31}
$$

From equation (30), we observe that the measurement noise is not completely removed. To effectively remove this measurement noise, we exploit the statistical properties of noise. The measurement noise follows a complex Gaussian distribution with zero mean and a variance of $\sigma^2(f)$ and its amplitude follows a Rayleigh distribution. Therefore, we design a Rayleigh filter to suppress measurement noise. The core of Rayleigh filter involves a weighted averaging of the entire signal to mitigate measurement noise. In this process, data points closer to the peak of the signal have a greater impact, receiving higher weights, while points farther away exert less influence and consequently receive reduced weights. The selection of weights for the Rayleigh filter is guided by the shape of the Rayleigh function. Due to the shape of the Rayleigh function, the peak points of the signal will be more strongly enhanced, while noise or low-amplitude signals will be suppressed. The one-dimensional Rayleigh probability density function is:

$$
f(x) = \frac{x}{\sigma^2}e^{-\frac{x^2}{2\sigma^2}}, x \geq 0, \tag{32}
$$

where x is the measurement noise amplitude, σ is the scale parameter of the Rayleigh function, which is related to the shape and peak position of the Rayleigh function. The Rayleigh function (32) illustrates a right-skewed waveform, with its peak occurring in the positive part. Hence, when designing the Rayleigh template, the size is usually considered to at least cover the peak area of the Rayleigh function. Specifically, when $x = \sigma$, the Rayleigh function value is maximum, so the size of the Rayleigh template we designed must be at least larger than σ. The size of the Rayleigh template designed in this article is $r\,(r > \sigma)$. To obtain specific weight coefficients, the Rayleigh function needs to be discretized. During discretization, the curve of the Rayleigh function can be sampled, and the obtained discrete Rayleigh function value can be used as the coefficient of the Rayleigh template. Subsequently, the mathematical expression of each weight coefficient value in the Rayleigh template is given by:

$$
w(i) = \frac{i-1}{\sigma^2}e^{-\frac{(i-1)^2}{2\sigma^2}}, i = 1, 2, ...r, \tag{33}
$$

where $w(i)$ is the i-th weight coefficient value in the template.The weights are normalized to sum to 1:

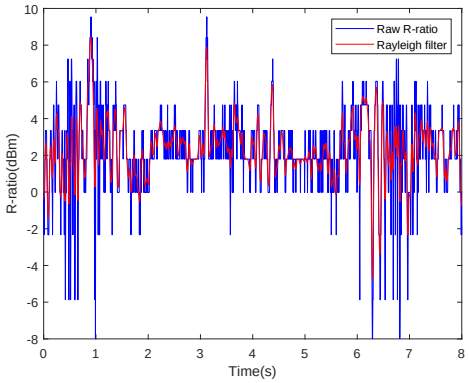


Fig. 7. R-ratio denoising.

$$normal_w\left(i\right)=\frac{w\left(i\right)}{\sum_{i}^{r}w\left(i\right)},i=1,2....r. \tag{34}$$

When applying this proposed Rayleigh template to the corresponding noise data, a convolution operation is usually used. The specific mathematical form is as follows:

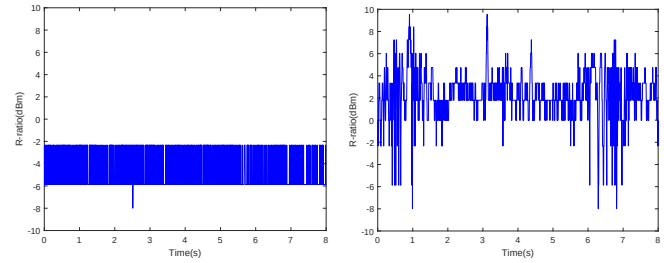
$$R\left(t\right)=\begin{bmatrix} R\left(m\right) \\ ... \\ R\left(t\right) \\ ... \\ R\left(n\right) \end{bmatrix}^{T}\begin{bmatrix} normal_w\left(1\right) \\ ... \\ normal_w\left(max_index\right) \\ ... \\ normal_w\left(r\right) \end{bmatrix}, \tag{35}$$

where $m=t-max_index+1$, $n=t+r-max_index$, and $max_index=\arg\max\left(\mathbf{normal_w}\right)$, representing the index of the maximum value in the weight. Fig. 7 illustrates the results of denoised R-ratio data through the proposed Rayleigh filter. Convolve the data to be processed with the designed Rayleigh template. And ensure that the key parts of the signal during the convolution process are aligned with the position of the maximum weight coefficient in the template. Through this process, we can successfully obtain denoised data.

B. Detection of sensing bounds

In our designed WiFi sensing system, the sensing bound represents the maximum distance at which intruders can be reliably detected. Beyond this distance, the system loses its effectiveness in sensing people. The magnitude of the sensing bound directly depends on the performance and sensitivity of the detection method. In this paper, we empoly customized FFT and wavelet transform method for intrusion detection to determine the sensing bound.

1) Improved FFT method based on maximum amplitude: In a typical indoor environment, due to multipath effects, environmental factors and noise interference, R-ratio may still exhibit fluctuation even in the absence of human movement. Fig. 8(a) illustrates the variation of R-ratio in a static environment, while Fig. 8(b) depicts the fluctuation of R-ratio caused by human intrusion. From Fig. 8, we can find different fluctuation patterns occur in static and intrusion environment. Particularly, during human movement, the signal waveform demonstrates rapid changes with specific frequency components. To deal with this challenge, we propose to use Fourier transform (FFT) to perform frequency domain analysis



(a) No-intruder scenario. (b) Intruder scenario.

Fig. 8. R-ratio variation.

on the R-ratio signal data. And we analyze the changes in R-ratio in the frequency domain to complete the detection of human intrusion motion. The WiFi works at 5.24 GHz and the sampling frequency is set to 200Hz, providing sufficient capability to capture the frequency changes associated with the human body movements.

Traditional methods typically emphasize the DC component of the signal, corresponding to the amplitude value at 0 Hz frequency. In signal processing, the 0-frequency component corresponds to the DC component of the signal, which is equal to the mean or offset value of the signal. The DC component usually represents the average level of the signal. However, in the intrusion scenario, we pay more attention to the dynamic changes of the signal caused by human movement, rather than solely on the overall average level. Therefore, we propose an improved FFT method that considers, the maximum amplitude except the 0-frequency component.

Specifically, we first apply fast Fourier transform to the R-ratio data. Since signals are real, and positive and negative frequencies are conjugate symmetric, we used half of the spectral length, as dropping the negative frequencies does not lead to any information loss. So we calculate the single-sided spectrum. The amplitudes in the original spectrum need to be scaled to maintain the total energy of the signal. Specifically, we double the amplitude of frequency components except the DC component and the Nyquist frequency.Then we choose maximum amplitude except the 0-frequency component as a key feature for intrusion detection. Similarly, we perform using fast Fourier transform on the static R-ratio data. Fig. 9 shows the single-sided spectrum in no-intruder scenario and intruder scenario. There is a significant difference between the maximum amplitude values except the 0-frequency component in no-intruder scenario and intruder scenario. We compare these two maximum amplitudes, and when the former is greater than the latter, it is considered that there is an intrusion.

2) Improved wavelet transform method based on scale projection: Since the human motion speed vary among individuals or for the same person at different times, we need a technique that can provide variable frequency resolution. Compared to other techniques, wavelet analysis provides both time and frequency locality at different resolutions as well as being computationally efficient.Hence, we utilize wavelet analysis for intrusion detection. In DWT, the generic step splits the signal into two parts: an approximation coefficient vector

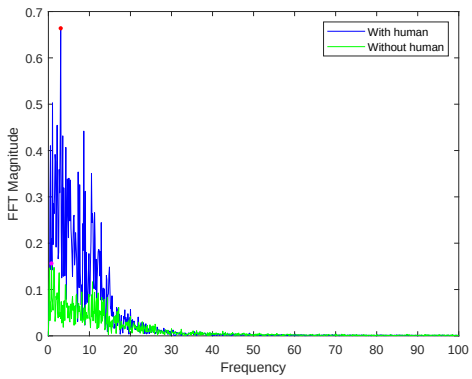


Fig. 9. Amplitude spectrum.

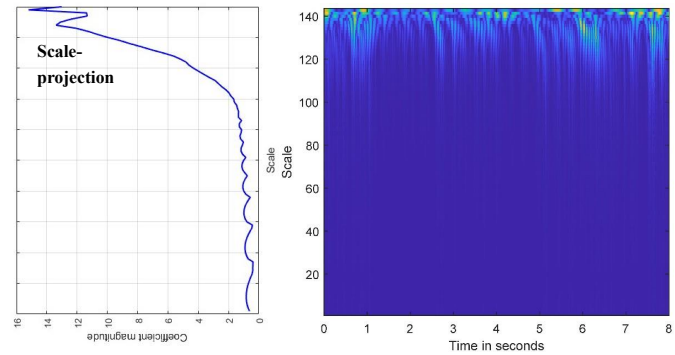
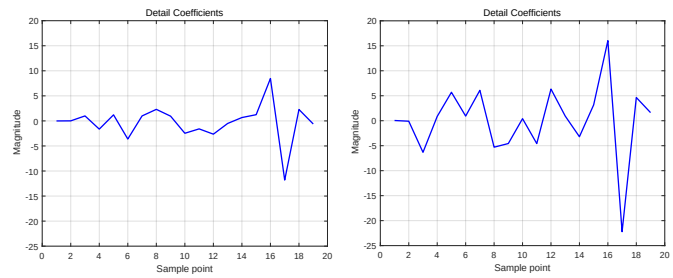


Fig. 10. Detecting frequencies of interest dynamically using scale projection.



(a) No-intruder scenario. (b) Intruder scenario.

Fig. 11. Detail coefficients.

and a detail coefficient vector. Detail coefficients typically contain local features in the signal which accurately capture characteristics associated with intrusion. Traditional wavelet transform use thresholding and other methods to process detail coefficients. However we focus on the fluctuation of detail coefficients. The level of decomposition in wavelet transform is a crucial parameter, determining the degree of signal decomposition in frequency and scale. As the level of decomposition increases, the level of the detail coefficients also increases accordingly. The detail coefficients of each level provide detailed information about the signal at different scales and frequencies. In intrusion detection, choosing the appropriate level of decomposition is crucial. Lower decomposition levels may fail to capture information at the frequency of human motion, while higher decomposition levels may bring noise or unnecessary details. Therefore, it is particularly necessary to select the level of decomposition based on the the frequency of human motion. In this paper, we proposes a method to identify the frequency of human motion based on scale projection. By accurately locating the main frequency of human motion, the system can more accurately capture signal characteristics related to intrusion events, thereby improving the accuracy and reliability of intrusion detection.

Specifically, to enhance the accuracy of intrusion detection, we leverage a wavelet-based denoising method for further refinement.Wavelet denoising consists of three stages: decomposition, thresholding detail coefficients, and reconstruction. In decomposition, DWT is recursively applied to break the signal into detail coefficients and approximation coefficients at different frequency levels. Thresholding is then applied to the wavelet detail coefficients to remove their noisy part. The threshold is dynamically selected, based on minimizing the Stein unbiased risk estimate. Finally, the denoised signal is reconstructed by combining the coefficients of the last approximation level with all thresholded details.

We experiment with scale projection to identify the main frequency we are interested in from the wavelet transform spectrogram, as shown in Fig. 10. The scale projection technique projects the spectrogram on the y-axis (i.e.the scale axis), maintaining the maximum value for each projected row. Then, it extracts the top local-maximums values:$\{(s_1, c_1), (s_2, c_2), (s_3, c_3), ..., (s_k, c_k)\}$, where (s_i, c_i) are the ith top scale and projected coefficient value at that scale, respectively.Finally, the scale of interest is estimated as:

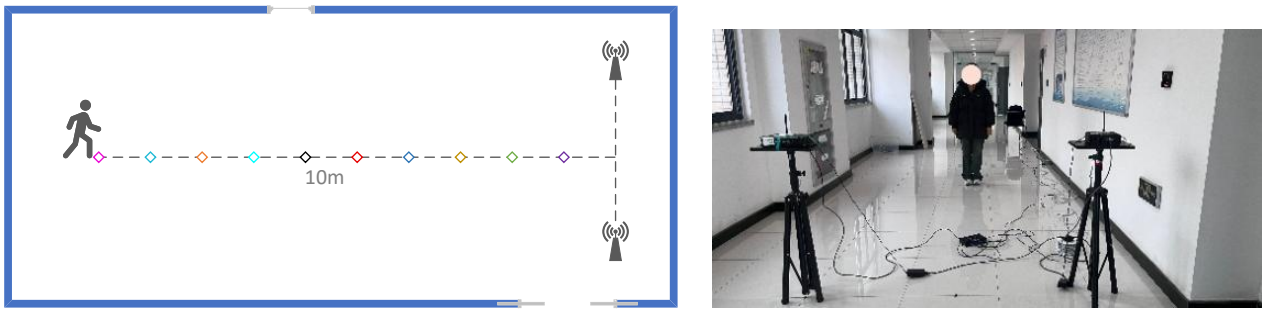
$$s^* = s\left(\arg\max\left(c_i\right)\right), s.t. i = 1, ...k. \tag{36}$$

After obtaining the scale information, according to the relationship between scale and frequency, we can get the frequency we are interested in:

$$f = \frac{Fs * wcf}{s^*}, \tag{37}$$

where Fs is sampling frequency, wcf is the center frequency of wavelet. Subsequently, relying on the identified frequency of interest, we can selectively choose an appropriate level of DWT decomposition to extract detail coefficients associated with specific motion frequencies. This process allows us to focus on signal features within the target frequency range, facilitating a more precise capture information relevant to motion events.

Fig. 11(a) is the detail coefficient waveform in a no-intruder scenario, and Fig. 11(b) depicts the detail coefficient waveform in an intruder scenario. We can clearly observe that in the presence of human movement, the fluctuation of the detail coefficient is significantly higher than in a no-intruder environment. Hence, we employ the method of calculating the variance of the detail coefficient for intrusion detection. By comparing the difference between the variance of the dynamic detail coefficient and the variance of the static detail coefficient, when the former is greater than the latter, an intrusion is considered to have occurred.



(a) Floorplan of the experiment setup.

(b) Positions of WiFi transceivers and the subject.

Fig. 12. Experiment scenario.

V. EVALUATION

A. Evaluation Setup

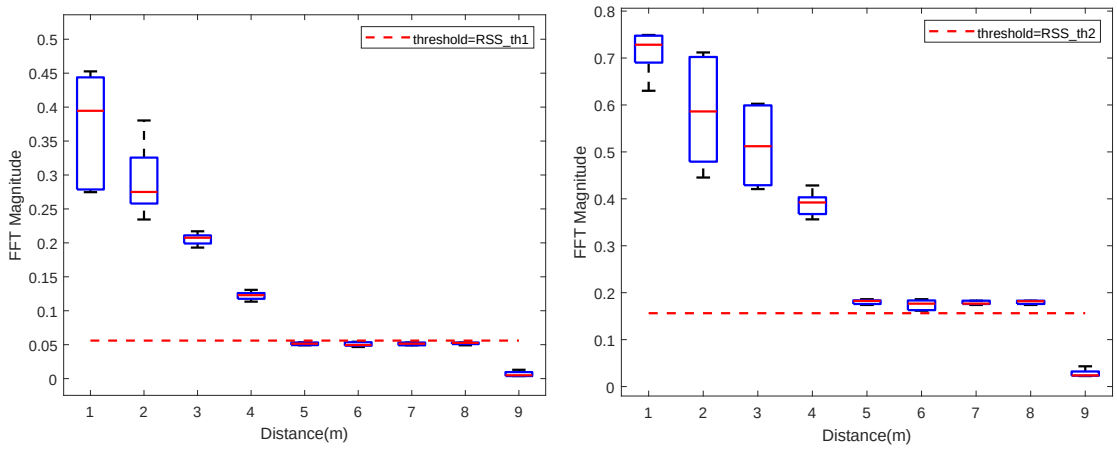
To evaluate the system's ability to extend bound, we conduct real experiments in a corridor. The corridor is 3 meters wide and 15 meters long. The experimental scenario is shown in Fig. 12, we place the Tx and Rx 1.2m apart. We decrease the distance between the volunteer and transceiver from 10 m to 1 m. At every 1 meter location, the volunteer walks back and forth for 8 seconds. In this experiment, we use commercial WiFi devices equipped with Intel 5300 NIC to collect RSS data, and each with one external antennas.The frequency is set to 5.24 GHz, bandwidth to 20 MHz and the transmitter sends 200 packets per second at a transmit power of 15dBm. To evaluate the system's intrusion detection range, we placed a series of tags on the ground to mark the distance from the WiFi link. During the experiment, we decrease the distance between the volunteer and transceiver from 10 m to 1 m. At every 1 meter location, the volunteer walks back and forth for 8 seconds.

B. Performance in intrusion detection

We first evaluate the improved FFT method based on the maximum amplitude. We first perform intrusion detection on the raw RSS data, and subsequently, we perform intrusion detection on the R-ratio data. In this article, we use the threshold method for intrusion detection. For the raw RSS data, the threshold (denoted as RSS_th1) is set as the maximum amplitude value, excluding the 0-frequency component processed by the improved FFT of the RSS data in a static environment. When the RSS data in the intrusion environment surpasses the threshold RSS_th1, it is interpreted as an intrusion. Similarly, for the R-ratio data, the threshold (denoted as RSS_th2) is established as the maximum amplitude value, excluding the 0-frequency component processed by the improved FFT of the R-ratio data in a static environment. Detection of intrusion occurs when the R-ratio data in the intrusion environment exceeds the threshold RSS_th2.

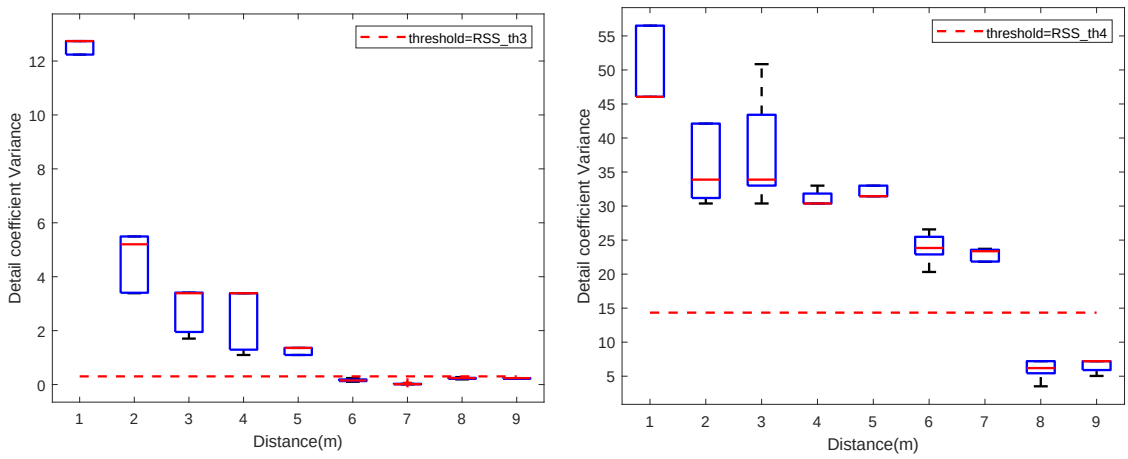
Fig. 13 illustrates the detection performance of the improved FFT method based on maximum amplitude, with Fig. 13(a) showcasing the results based on raw RSS data and Fig. 13(b) presenting the esults based on R-ratio. We can observe that as the distance increases, there is a gradual decrease in FFT Magnitude. In the case of raw data (Fig. 13(a)), when the distance extends to 5 meters, the FFT Magnitude falls below the threshold RSS_th1, establishing the perceptual bound at 4 meters. Conversely, for R-ratio data (Fig. 13(b)), when the distance reaches 9 meters, the FFT Magnitude drops below the threshold RSS_th2, resulting in a sensing bound of 8 meters. This observation shows that the data processing method based on R-ratio has the potential to double the sensing range compared to raw RSS data, highlighting its efficacy in enhancing detection performance at greater distances.

Then, we evaluate the wavelet transform method based on scale projection. We first perform intrusion detection on the raw RSS data, and subsequently, we perform intrusion detection on the R-ratio data. Similar to the previous method, the threshold method is employed for intrusion detection. For the raw RSS data, the threshold (denoted as RSS_th3) is determined as the variance of the detail coefficient processed by the improved wavelet transform of the RSS data in a static environment. When the RSS data in the intrusion environment surpasses the threshold RSS_th3, it is interpreted as an intrusion. Similarly, for R-ratio data, the threshold (denoted as RSS_th4) is established as the variance of the detail coefficient processed by the improved wavelet transform of the R-ratio data in a static environment. Detection of intrusion occurs when the R-ratio data in the intrusion environment exceeds the threshold RSS_th4. Fig. 14 illustrates the detection performance of the improved wavelet transform method based on scale projection, with Fig. 14(a) presenting the results based on raw RSS data and Fig. 14(b) showcasing outcomes based on R-ratio. We can observe that as the distance increases, there is a gradual decrease in the variance of the detail coefficient. For raw data (Fig. 14(a)), when the distance extends to 6 meters, the variance of the detail coefficient falls below the threshold, establishing the sensing bound at 5 meters, as depicted in the figure. Similarly, for R-ratio data (Fig. 14(b)), when the distance reaches 8 meters, the variance of the detail coefficient drops below the threshold, resulting in a sensing bound of 7 meters. Combined with Fig. 13(a) and Fig. 13(b), experiments results show that the data processing method based on R-ratio can extend the sensing range.



(a) Detection performance based on raw data. (b) Detection performance based on R-ratio.

Fig. 13. Improved FFT method based on maximum amplitude.



(a) Detection performance based on raw data. (b) Detection performance based on R-ratio.

Fig. 14. Improved wavelet transform method based on scale projection.

C. Comparison with existing work

In this section, we compare our work with existing research on motion detection systems. Specifically, we analyze the differences between R-ratio-based FFT, R-ratio-based DWT, and three RSS-based motion detection systems: WiGest [25], HAR-FFT [27], and Simple-HomeFi [8]. In Simple-HomeFi, only the variance calculation operation is used.

From Fig. 15, we observe that when the volunteer is 3 meters away from the WiFi transceiver device, all methods achieve a 100% detection rate. Even when the volunteer is 5 meters away, WiGest, R-ratio-based FFT, and R-ratio-based DWT still maintain high detection rates. However, when the volunteer moves further away to a distance of 8 meters, only R-ratio-based FFT achieves a 100% detection rate. These results demonstrate the superior performance of our approach in detecting intrusions at long distances. Compared to WiGest and other methods, our approach enables higher detection accuracy at longer distances.

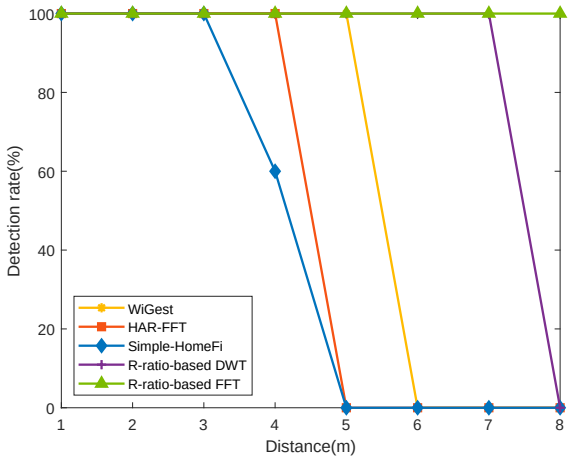


Fig. 15. Comparison between different methods.

D. Impact of intruders with different body sizes

From equation (9), it becomes evident that human body sizes play a significant role in influencing RSS, consequently

TABLE I. Body size parameter table of different volunteers

Volunteers	Gender	Height(cm)	Width(cm)	Length(cm)
1	male	178	25	45
2	male	178	25	40
3	male	170	20	40
4	female	168	15	30
5	female	160	15	30
6	female	155	15	25

impacting the sensing bound. In our experiment, we purposefully recruited six volunteers with diverse body sizes, comprising three men and three women. The body size details of these participants are presented in Table I. We use the improved FFT method based on maximum amplitude for intrusion detection.

As illustrated in Fig. 16, with an increase in detection distance, the FFT Magnitude of each volunteer exhibits a decline. This phenomenon is attributed to indoor signal propagation losses caused by signal attenuation and multipath effects. In addition, larger-sized volunteers tend to exhibit higher FFT Magnitude. For example, Volunteer 1, being the largest, reflects a larger area, leading to a more extensive blocked area and a more pronounced impact on RSS. Consequently, intrusion motion can be detected at a position as far as 8 meters. In contrast, Volunteer 6, with the smallest body size, reflects a relatively smaller signal area, resulting in a less conspicuous impact on RSS. As a result, intrusion motion can only be detected at a position of 6 meters.

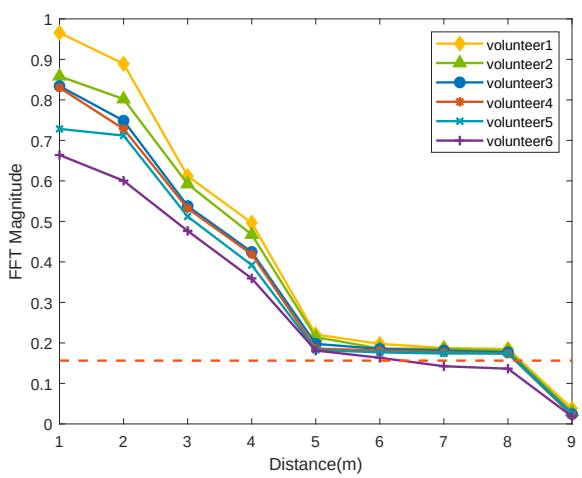


Fig. 16. Comparison between volunteers with different body sizes.

According to the equations (11) and (16), both the blocked area and reflected area are intricately linked to human body size, exerting a direct influence on RSS. Consequently, when developing an intrusion detection system, careful consideration must be given to the placement of the WiFi transceiver. Optimal placement entails minimizing the blocked area while maximizing the reflected area. This decision involves factors such as choosing the height, angle, and orientation of the antenna, as well as formulating an overall placement strategy for the device. In practical applications, system parameters can be adjusted, such as dynamically adjusting the threshold, to adapt to intruders of various sizes.

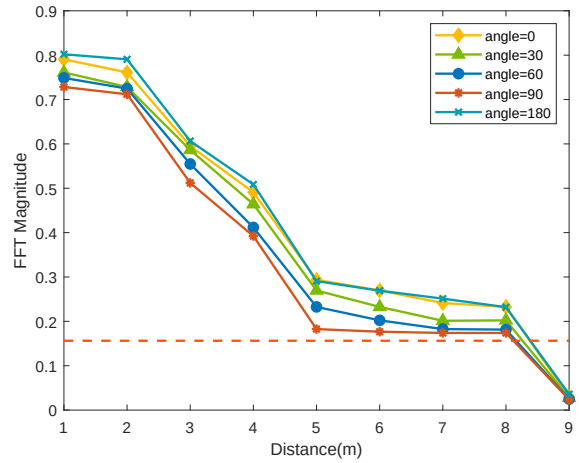


Fig. 17. Comparison between different angles.

E. Impact of angles

According to equation (10), the angle between the person and the WiFi link also has a significant impact on RSS. Therefore, in our experiment, we invited one of the volunteers to pose at various angles: 0 degrees (directly facing the WiFi link), 30 degrees, 60 degrees, 90 degrees, and 180 degrees(facing away from the WiFi link). To perform intrusion detection, We use the improved FFT method based on maximum amplitude. As depicted in Fig. 17, there is a noticeable decrease in FFT Magnitude at each angle as the detection distance increases. And larger angles correspond to smaller FFT Magnitudes. The impact of volunteers on RSS is more pronounced at 0 degrees compared to 90 degrees, primarily because, at 0 degrees, both the reflected and blocked areas of the human body are larger, leading to a more conspicuous effect on RSS. Interestingly, the effects at 0 degrees and 180 degrees are nearly identical due to the equal front and back areas of the human body, resulting in similar reflected and blocked areas. The experiment revealed that when a person moves directly towards the WiFi link (0 degrees), the reflected and blocked areas of the human body are more extensive, significantly influencing RSS. Consequently, in the design of an intrusion detection system, strategic placement of the WiFi transceiver facing people can be considered to amplify changes in RSS signals, facilitating the detection of intrusion events. In practical applications, the parameters of the intrusion detection algorithm can be fine-tuned to specific needs, including dynamic adjustments of the RSS threshold to accommodate intrusion behaviors from different angles effectively.

Overall, the derived equation offers valuable insights and initial guidance regarding the influence of intruder size and angle on RSS. However, the precise placement of WiFi transceivers and the fine-tuning of system parameters demand more comprehensive and in-depth research and verification. Future research can further optimize the model and algorithm based on this paper to better adapt to the needs of intrusion detection under various environmental conditions.

VI. Conclusion

In this article, to address the limitations of existing RSS-based intrusion detection solutions, we introduced a novel model of motion-disturbed RSS for improving the accuracy of the WiFi sensing bound. The proposed R-ratio metric served as an effective performance measure to extend this bound, taking into account the blocked and reflection areas within the motion-disturbed RSS model. Additionally, an efficient noise filter was designed to further enhance the system's robustness. We also provided two customized intrusion detection methods to evaluate the R-ratio performance. The experimental results demonstrated the superiority of the R-ratio-based intrusion detection method, showcasing an approximate doubling of the WiFi sensing bound compared to other approaches utilizing raw RSS data. Furthermore, the proposed motion-disturbed RSS model offered valuable insights and guidance for enhancing the overall performance of intrusion detection systems.

In the future, we can further optimize our proposed motion-disturbed RSS model to better adapt to the needs of intrusion detection under various environmental conditions. Based on the our proposed motion-disturbed RSS model and R-ratio metric, future research can further explore model optimization and algorithms in various aspects such as the layout of WiFi transceivers and the adjustment of system parameters.

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